

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following questions.

1. State the principle of virtual work and the D' Alembert's principle accompanied with brief explanations. (5x8=40)

2. Use D' Alembert's principle to derive Euler-Lagrange equations of motion

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} = Q_i$$

3. An ant crawls on the surface of a ball of radius b in such a way that the ant's motion is given in spherical coordinates

$$r = b, \quad \theta = \frac{\pi}{2} \left[1 + \frac{1}{4} \cos 4\omega t \right], \quad \phi = \omega t$$

Find the speed of the ant as a function of time t .

4. What are Lagrange multipliers? Use the method of Lagrange multiplier to derive the Lagrange equations of motion for nonholonomic constraints.

5. Consider a cylinder rolling down an inclined plane. Write the equation of constraint. Use the method of Lagrange multiplier to write down the equation of motion.

Q.2. Solve the following questions.

1. Show that the functional integral

$$I = \int_{x_1}^{x_2} f[y(x), y'(x), x] dx,$$

will be minimum only when Euler's equation is satisfied i.e.

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0.$$

Consider Coulomb scattering of charged particles. The Coulomb repulsive potential $V(r) = \frac{k}{r}$. Start with the equation

$$\theta(r) = \int \frac{l dr}{r^2 \sqrt{2\mu \left(E - V - \frac{l^2}{2\mu r^2} \right)}} + \text{constant},$$

by making suitable substitutions, derive the following expression of the

$$b = \frac{k}{2E} \cot \left(\frac{\varphi}{2} \right)$$

φ is the scattering angle and E is the total energy of the particle.

Now that the transformation

$$q = PQ^2,$$

$$p = \frac{1}{Q},$$

is canonical and also show that the corresponding generating function is

$$F = \frac{1}{Q}$$

D'ALEMBERT'S PRINCIPLE

In the principle of virtual work, coefficients of $\delta \vec{r}_i$ are not equal to zero, in general. To make coefficients zero, transform the principle of virtual work into displacement of q_i . This principle only deals with statics but a general principle of motion of the system is needed. To derive such a principle, again start from equation of motion. From Newton's second law of motion:

$$\vec{F}_i = \vec{\dot{p}}_i$$

$$\vec{F}_i - \vec{\dot{p}}_i = 0$$

Thus

$$\sum \vec{F} = \vec{F}_i + (-\vec{\dot{p}}_i) = 0$$

Therefore a system will be in equilibrium under a force equal to sum of actual force ($\vec{F}_i$) and reversed effective force ($-\vec{\dot{p}}_i$). So principle of virtual work becomes

$$\sum_i (\vec{F}_i - \vec{\dot{p}}_i) \cdot \delta \vec{r}_i = 0$$

$$\sum_i (\vec{F}_i^{(a)} + \vec{f}_i - \vec{\dot{p}}_i) \cdot \delta \vec{r}_i = 0 \quad \because (\vec{F}_i = \vec{F}_i^{(a)} + \vec{f}_i)$$

$$\sum_i (\vec{F}_i^{(a)} - \vec{\dot{p}}_i) \cdot \delta \vec{r}_i + \sum_i \vec{f}_i \cdot \delta \vec{r}_i = 0$$

Again virtual work done due to constraint force vanishes. i.e.,

$$\sum_i \vec{f}_i \cdot \delta \vec{r}_i = 0$$

So above equation becomes

$$\sum_i (\vec{F}_i^{(a)} - \vec{\dot{p}}_i) \cdot \delta \vec{r}_i = 0$$

This is called D'Alembert's principle.

PRINCIPLE OF VIRTUAL WORK

An arbitrary infinitesimal change of coordinates δr_i of a system while time is held constant is called virtual displacement.

Consider a system of particles having n independent generalized coordinates at a given time t . Then the translation vector will be

$$\vec{r}_i = \vec{r}_i(q_1, q_2, \dots, q_n, t)$$

Using chain rule of partial differentiation δr_i becomes

$$\delta \vec{r}_i = \frac{\partial \vec{r}_i}{\partial q_1} \delta q_1 + \frac{\partial \vec{r}_i}{\partial q_2} \delta q_2 + \dots + \frac{\partial \vec{r}_i}{\partial q_n} \delta q_n + \frac{\partial \vec{r}_i}{\partial t} \delta t$$

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j + \frac{\partial \vec{r}_i}{\partial t} \delta t$$

The virtual displacement is quite different from real displacement as no real displacement can take place without the passage of time. i.e., $\delta t = 0$. So virtual displacement becomes

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j + \frac{\partial \vec{r}_i}{\partial t} (0)$$

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CLASSICAL MECHANICS

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j + 0$$

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j$$

This is called virtual displacement.

If the system is in equilibrium at some time t then

$$\vec{F}_i = 0$$

So the virtual work of the system will be

$$\sum_i \vec{F}_i \cdot \delta \vec{r}_i = 0$$

As force $\vec{F}_i$ is effective force which is the sum of applied force $\vec{F}_i^{(a)}$ and constraint force $\vec{f}_i$, i.e.,

$$\vec{F}_i = \vec{F}_i^{(a)} + \vec{f}_i$$

The virtual work done, now, becomes

$$\sum_i (\vec{F}_i^{(a)} + \vec{f}_i) \cdot \delta \vec{r}_i = 0$$

$$\sum_i \vec{F}_i^{(a)} \cdot \delta \vec{r}_i + \sum_i \vec{f}_i \cdot \delta \vec{r}_i = 0$$

For simplicity, consider the virtual work done by constraint force is zero (for instance rigid bodies). i.e., $\sum_i \vec{f}_i \cdot \delta \vec{r}_i = 0$

So above equation becomes

$$\sum_i \vec{F}_i^{(a)} \cdot \delta \vec{r}_i = 0$$

This is called the principle of virtual work.

In above equation the dot product of two vectors is zero. This means that the coefficient of $\delta \vec{r}_i$ not necessarily to be zero. i.e., in general $\vec{F}_i^{(a)} \neq 0$.

Virtual displacement is

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j \quad \text{----- (1)}$$

Velocity $\vec{V}_i$ is

$$\vec{V}_i = \frac{d\vec{r}_i}{dt}$$

Using chain rule of partial differentiation for $x(t)$ and $y(t)$

$$\frac{df(x, y)}{dt} = \left(\frac{\partial f}{\partial x} \right) \left(\frac{dx}{dt} \right) + \left(\frac{\partial f}{\partial y} \right) \left(\frac{dy}{dt} \right)$$

Velocity $\vec{V}_i$ becomes

$$\vec{V}_i = \left(\frac{\partial \vec{r}_i}{\partial q_1} \right) \left(\frac{dq_1}{dt} \right) + \left(\frac{\partial \vec{r}_i}{\partial q_2} \right) \left(\frac{dq_2}{dt} \right) + \dots + \left(\frac{\partial \vec{r}_i}{\partial q_n} \right) \left(\frac{dq_n}{dt} \right) + \left(\frac{\partial \vec{r}_i}{\partial t} \right) \left(\frac{dt}{dt} \right)$$

$$\vec{V}_i = \frac{\partial \vec{r}_i}{\partial q_1} \dot{q}_1 + \frac{\partial \vec{r}_i}{\partial q_2} \dot{q}_2 + \dots + \frac{\partial \vec{r}_i}{\partial q_n} \dot{q}_n + \frac{\partial \vec{r}_i}{\partial t}$$

$$\vec{V}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_i}{\partial t} \quad \text{----- (2)}$$

Virtual work done in terms of generalized coordinates is

$$W = \sum_i \vec{F}_i \cdot \delta \vec{r}_i$$

Using value of virtual displacement from eq(1)

$$W = \sum_i \vec{F}_i \cdot \delta \vec{r}_i \quad W = \sum_i \vec{F}_i \cdot \left(\sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j \right) \quad \therefore \delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j$$

$$W = \sum_{i,j} \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j$$

$$W = \sum_j Q_j \delta q_j \quad \text{----- (3)}$$

Where Q_j is called **generalized force** and given by

$$Q_j = \sum_i \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \quad \text{----- (4)}$$

D'Alembert's principle is

$$\sum_i (\vec{F}_i - \vec{p}_i) \cdot \delta \vec{r}_i = 0$$

$$\sum_i \vec{F}_i \cdot \delta \vec{r}_i - \sum_i \vec{p}_i \cdot \delta \vec{r}_i = 0$$

Using eq(1) & eq(3)

$$\sum_j Q_j \delta q_j - \sum_{i,j} m_i \frac{d\vec{V}_i}{dt} \cdot \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j = 0$$

$$\sum_j Q_j \delta q_j - \sum_{i,j} m_i \left[\frac{d}{dt} \left(\vec{V}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \right) - \vec{V}_i \cdot \frac{d}{dt} \left(\frac{\partial \vec{r}_i}{\partial q_j} \right) \right] \delta q_j = 0$$

From eq(2), $\frac{\partial \vec{V}_i}{\partial \dot{q}_j} = \frac{\partial \vec{r}_i}{\partial q_j}$, so above equation becomes

$$\sum_j Q_j \delta q_j - \sum_{i,j} m_i \left[\frac{d}{dt} \left(\vec{V}_i \cdot \frac{\partial \vec{V}_i}{\partial \dot{q}_j} \right) - \vec{V}_i \cdot \frac{\partial \vec{V}_i}{\partial q_j} \right] \delta q_j = 0$$

$$\sum_j Q_j \delta q_j - \sum_{i,j} m_i \left[\frac{d}{dt} \left\{ \frac{\partial}{\partial \dot{q}_j} \left(\frac{1}{2} \vec{V}_i^2 \right) \right\} - \frac{\partial}{\partial q_j} \left(\frac{1}{2} \vec{V}_i^2 \right) \right] \delta q_j = 0$$

$$\sum_j Q_j \delta q_j - \sum_j \left[\frac{d}{dt} \left\{ \frac{\partial}{\partial \dot{q}_j} \left(\sum_i \frac{1}{2} m_i \vec{V}_i^2 \right) \right\} - \frac{\partial}{\partial q_j} \left(\sum_i \frac{1}{2} m_i \vec{V}_i^2 \right) \right] \delta q_j = 0$$

$$\sum_j \left[\frac{d}{dt} \left\{ \frac{\partial}{\partial \dot{q}_j} \left(\sum_i \frac{1}{2} m_i \vec{V}_i^2 \right) \right\} - \frac{\partial}{\partial q_j} \left(\sum_i \frac{1}{2} m_i \vec{V}_i^2 \right) - Q_j \right] \delta q_j = 0$$

$$\sum_j \left\{ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j \right\} \delta q_j = 0$$

★ Where $T \equiv \sum_i \frac{1}{2} m_i \vec{V}_i^2$ is called kinetic energy.

If the constraints are holonomic then there is a set of independent coordinates q_j that contain the constraint conditions implicitly in the transformation equations. For any virtual displacement δq_j to be independent of another virtual displacement δq_k above equation can be written as

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j = 0$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j \quad \text{For dynamic} \quad \text{----- (5)}$$

This is called Nielsen's form of Lagrange equation.

An ant crawls on the surface of a ball of radius a in such a manner that the ant's motion is given in spherical coordinates by the equations

$$r = a, \phi = 2\omega t, \theta = \frac{\pi}{2} \left(1 + \frac{1}{4} \cos(4\omega t) \right)$$

- Find the speed of the ant as a function of time t .
- Determine the acceleration of the ant as a function of time t .

Spherical Coordinate System

In spherical coordinate system, the three coordinates: distance (r) from a fixed point (origin), polar angle (θ), and azimuthal angle (ϕ), determine the position of any point. The position, velocity and

acceleration can be written as, $\vec{r} = r\hat{r}$, $\vec{v} = \frac{d\vec{r}}{dt}$ and $\vec{a} = \frac{d\vec{v}}{dt}$ respectively.

Answer and Explanation: 1

Given Data

- The equations for the motion of the ant are

$$r = a, \phi = 2\omega t, \theta = \frac{\pi}{2} \left(1 + \frac{1}{4} \cos(4\omega t) \right).$$

(a)

The formula to find the velocity in spherical coordinates is expressed below:

$$\vec{v} = \dot{r}\hat{r} + r\dot{\phi}\sin\theta\hat{\phi} + r\dot{\theta}\hat{\theta} \quad (1)$$

From the given equations of motion,

$$\dot{r} = 0$$

$$\dot{\phi} = 2\omega$$

$$\dot{\theta} = -\frac{\pi\omega}{2}\sin 4\omega t$$

Inserting all these values in equation (1), we get the following:

$$\vec{v} = 0\hat{r} + a(2\omega)\sin\left(\frac{\pi}{2}\left(1 + \frac{1}{4}\cos(4\omega t)\right)\right)\hat{\phi} - a\left(\frac{\pi\omega}{2}\right)\sin(4\omega t)\hat{\theta}$$

Now, using the trigonometric identity,

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

The velocity of the ant is,

$$\begin{aligned}\vec{v} &= 2a\omega\left(\sin\frac{\pi}{2}\cos\left(\frac{\pi}{8}\cos 4\omega t\right) + \cos\frac{\pi}{2}\sin\left(\frac{\pi}{8}\cos 4\omega t\right)\right)\hat{\phi} - \frac{\pi a\omega}{2}\sin 4\omega t\hat{\theta} \\ &= 2a\omega\cos\left(\frac{\pi}{8}\cos 4\omega t\right)\hat{\phi} - \frac{\pi a\omega}{2}\sin 4\omega t\hat{\theta}\end{aligned}$$

The speed of the ant is,

$$|\vec{v}| = 2a\omega\sqrt{\cos^2\left(\frac{\pi}{8}\cos 4\omega t\right) + \frac{\pi^2}{16}\sin^2 4\omega t}$$

Thus, the speed of the ant is

$$|\vec{v}| = 2a\omega\sqrt{\cos^2\left(\frac{\pi}{8}\cos 4\omega t\right) + \frac{\pi^2}{16}\sin^2 4\omega t}.$$

$$\Rightarrow \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) + \lambda \frac{\partial g}{\partial y} = 0 \quad (34)$$

$$\text{and } \frac{\partial f}{\partial z} - \frac{d}{dx} \left(\frac{\partial f}{\partial z'} \right) + \lambda \frac{\partial g}{\partial z} = 0 \quad (35)$$

Generalizing we get

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial f}{\partial y_i'} \right) + \sum_{k=1}^m \lambda_k(x) \frac{\partial g_k}{\partial y_i} = 0 \quad (36)$$

$i = 1, 2, \dots, n$ and $k = 1, 2, \dots, m$

The functions $\lambda_k(x)$ are called Lagrange multipliers

The Lagrange Equation of motion is given by

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} = Q_i \quad (37)$$

If there are some constraints then

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} = Q_i + Q_i' \quad (38)$$

where Q_i' are the forces of constraints.

For a Conservative system Lagrange Equation in association with constraints is given by

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i \quad (39)$$

$$\text{where } Q_i' = \sum_{k=1}^m \lambda_k(t) \frac{\partial g_k}{\partial q_i} \quad (40)$$

So the general equation for constraints is

$$g_k(q_1, q_2, \dots, q_n, t) = 0$$

Upto Now our whole discussion was about holonomic constraints. Now we discuss the case for non holonomic constraints.

Non Holonomic Constraint System: The

equation for non holonomic constraints is given by

$$\sum_{i=1}^n a_{ki} dq_i + a_k dt = 0 \quad (41)$$

$i = 1, \dots, n, k = 1, 2, \dots, m$

Equation (41) for virtual displacement is given by

$$\sum_i a_{ki} \delta q_i = 0 \quad (42)$$

because for virtual displacement $\delta t = 0$

multiplying (42) by $\lambda(t)$

$$\lambda(t) \sum_i a_{ki} \delta q_i = 0 \quad (43)$$

So for the system of non-holonomic constraints

(39) becomes

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \sum_{k=1}^m \lambda_k(t) a_{ki} \delta q_i \quad (44)$$

Now in order to find Lagrange multipliers

$$\text{but } \sum_{k=1}^m \lambda_k(t) a_{ki} \delta q_i = 0 \quad (45)$$

THE EULER-LAGRANGE EQUATION: From fig (c) if

- we make an infinitesimal displacement of "x" from x_0 then
- it leaves $f(x)$ unchanged.
- Now Consider two points $A(x_1, y_1)$ and $B(x_2, y_2)$ in fig (d)
- We are to find the function $y(x)$ for which the path between A and B is minimum. As stated above that in elementary calculus any infinitesimal displacement of x from a stationary point does not affect the function $f(x)$ similarly we introduce a small variation in actual path.

Our functional is given by

$$I[y] = \int_{x_1}^{x_2} f[y(x), y'(x), x] dx$$

If we will introduce a variation then the integral evaluated for actual path is less than for any neighbouring path. Let $\eta(x)$ be small variation then we may write $y_\alpha(x) = y(x) + \alpha \eta(x)$ where α is a parameter. If $\alpha=0$ the $y_\alpha(x)$ is the actual path.

Also $y_\alpha(x_1) = 0$ and $y_\alpha(x_2) = 0$ — 3(a,b)

$$I[y_\alpha] = \int_{x_1}^{x_2} f[y_\alpha(x), y'_\alpha(x), x] dx$$

So we may say that $I[y_\alpha]$ has minimum value for $\alpha=0$ hence

$$\frac{dI}{d\alpha} = 0 \quad \text{at } \alpha=0$$

$$\frac{dI[y_\alpha]}{d\alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y_\alpha} \frac{\partial y_\alpha}{\partial \alpha} + \frac{\partial f}{\partial y'_\alpha} \frac{\partial y'_\alpha}{\partial \alpha} + \frac{\partial f}{\partial x} \frac{\partial x}{\partial \alpha} \right) dx$$

$$= \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y_\alpha} \frac{\partial y_\alpha}{\partial \alpha} + \frac{\partial f}{\partial y'_\alpha} \frac{\partial y'_\alpha}{\partial \alpha} + 0 \right) dx = \frac{\partial I}{\partial \alpha} = 0$$

From (2) $\frac{\partial y_\alpha}{\partial \alpha} = \eta(x)$ $y_\alpha(x) = y(x) + \alpha \eta(x)$

Also $y'_\alpha = \alpha \eta'(x) + y'(x)$ $\frac{\partial y_\alpha}{\partial \alpha} = \eta(x)$

$\frac{\partial y'_\alpha}{\partial \alpha} = \eta'(x)$ $\frac{\partial y'_\alpha}{\partial \alpha} = \eta'(x)$

So above equation becomes

$$\frac{dI[y_\alpha]}{d\alpha} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y_\alpha} \eta(x) + \frac{\partial f}{\partial y'_\alpha} \eta'(x) \right) dx \quad \text{--- (4)}$$

$$\frac{dI[y_\alpha]}{d\alpha} = \int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta(x) dx + \int_{x_1}^{x_2} \frac{\partial f}{\partial y'_\alpha} \eta'(x) dx \quad \text{--- (4a)}$$

Now taking

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y'_\alpha} \eta'(x) dx$$

$$d\alpha = x_1 \int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta(x) dx + x_2 \int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta(x) dx \quad (4a)$$

Now taking $\int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta'(x) dx$

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(4)

Integrating by parts.

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta'(x) dx = \left[\frac{\partial f}{\partial y_\alpha} \eta(x) \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial f}{\partial y_\alpha} \right) \eta(x) dx$$

$$\int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta'(x) dx = - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial f}{\partial y_\alpha} \right) \eta(x) dx \quad (5)$$

put in (4a)

$$\int_{x_1}^{x_2} \frac{dI[y_\alpha]}{d\alpha} = \int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha} \eta(x) dx - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial f}{\partial y_\alpha} \right) \eta(x) dx$$

$$\frac{dI[y_\alpha]}{d\alpha} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y_\alpha} - \frac{d}{dx} \left(\frac{\partial f}{\partial y_\alpha} \right) \right] \eta(x) dx$$

Adding Condition $\frac{dI[y_\alpha]}{d\alpha} \Big|_{\alpha=0} = 0$

$$\Rightarrow \left(\frac{\partial f}{\partial y_\alpha} - \frac{d}{dx} \left(\frac{\partial f}{\partial y_\alpha} \right) \right) = 0 \quad (6)$$

This equation (6) is called Euler-Lagrange Equation.

So the functional $I[y]$ has least value if $f=y(x)$ satisfies Euler-Lagrange Equation.

STEPS FOR USING CALCULUS OF VARIATIONS

and answer of each exercise is given in relevant section of chapter)

Ex. 4.1:- Consider Coulomb scattering of charged particles. The Coulomb repulsive potential is,

$$V = \frac{k}{r}$$

Start with equation

$$\theta(r) = \int \frac{l dr}{r^2 \sqrt{2\mu \left(E - V - \frac{l^2}{\mu^2 r^2} \right)}}$$

By making suitable substitutions, derive following expression for impact parameter,

$$b = \frac{k}{2E} \cot \left(\frac{\varphi}{2} \right)$$

φ is scattering angle and E is total energy of particle. Also show that differential cross-section of scattering is given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{4} \left(\frac{k}{2E} \right)^2 \frac{1}{\sin^2 \left(\frac{\varphi}{2} \right)}$$

Answer:- Let

$$r = \frac{1}{u}, \text{ then } V = \frac{k}{r} = ku \text{ and } \theta(r) = \int \frac{l \times -r^2 du}{r^2 \sqrt{2\mu \left(E - ku - \frac{l^2 u^2}{\mu^2} \right)}}$$

$$\Rightarrow \theta(r) = - \int \frac{l du}{\sqrt{2\mu \left(E - ku - \frac{l^2 u^2}{\mu^2} \right)}} = - \int \frac{l du}{\sqrt{- \left(u^2 + \frac{2\mu k}{l^2} u - \frac{2\mu E}{l^2} \right)}}$$

Let

$$\frac{\mu k}{l^2} = \alpha \text{ and } \beta = \frac{2\mu E}{l^2}, \text{ then}$$

$$\theta(r) = - \int \frac{ldu}{\sqrt{-\{(u + \alpha)^2 - (\alpha^2 + \beta^2)\}}} = \int \frac{ldu}{\sqrt{(\sqrt{\alpha^2 + \beta^2})^2 - (u + \alpha)^2}}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{u + \alpha}{\sqrt{\alpha^2 + \beta^2}} \right) \Rightarrow \cos \theta = \frac{u + \frac{\mu k}{l^2}}{\sqrt{\alpha^2 + \beta^2}} \Rightarrow \cos \theta = \frac{\frac{u}{\alpha} + 1}{\sqrt{1 + \frac{\beta^2}{\alpha^2}}}$$

$$\Rightarrow \sqrt{1 + \frac{\beta^2}{\alpha^2}} \cos \theta = \frac{u}{\alpha} + 1 \Rightarrow e \cos \theta = \frac{u}{\alpha} + 1 \text{ where } e = \sqrt{1 + \frac{\beta^2}{\alpha^2}} = \sqrt{1 + \frac{2El^2}{\mu k^2}}$$

Since $E > 0$, so $e > 1$, which shows that equation represents a hyperbola.

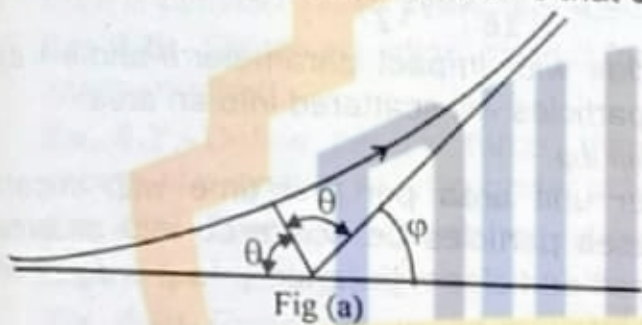


Fig (a)



Fig (b)

The angle between incident ray and scattered ray is called angle of deflection. From equation,

$$e \cos \theta = \frac{u}{\alpha} + 1, \text{ we have } e \cos \theta = \frac{1}{r\alpha} + 1 \Rightarrow r = \frac{1/\alpha}{-1 + e \cos \theta}$$

When $r \rightarrow \infty$, $\theta = \theta_m$, $-1 + e \cos \theta_m = \frac{1}{\infty} \Rightarrow \theta_m = \cos^{-1} \left(\frac{1}{e} \right)$. For large r , from fig (a),

$$\varphi = \pi - 2\theta_m \Rightarrow 2\theta_m = \pi - \varphi \Rightarrow 2\cos^{-1} \left(\frac{1}{e} \right) = \pi - \varphi \Rightarrow \cos^{-1} \left(\frac{1}{e} \right) = \frac{\pi}{2} - \frac{\varphi}{2}$$

$$\Rightarrow \frac{1}{e} = \cos \left(\frac{\pi}{2} - \frac{\varphi}{2} \right) \Rightarrow \frac{1}{e} = \sin \left(\frac{\varphi}{2} \right) \Rightarrow \frac{1}{e^2} = \sin^2 \left(\frac{\varphi}{2} \right) \Rightarrow \csc^2 \left(\frac{\varphi}{2} \right) = e^2$$

$$\Rightarrow \csc^2 \left(\frac{\varphi}{2} \right) = 1 + \frac{2El^2}{\mu k^2} \Rightarrow \csc^2 \left(\frac{\varphi}{2} \right) - 1 = \frac{2El^2}{\mu k^2} \Rightarrow \cot^2 \left(\frac{\varphi}{2} \right) = \frac{2El^2}{\mu k^2} \rightarrow (a)$$

Solid angle: The angle subtended by an area at a point not in its plane is called solid angle. It is usually denoted by $d\Omega = dA/r^2$ and $\Omega = 4\pi$.

Impact parameter: The perpendicular distance between the line of incoming beam and the line passing through the scatterer is called impact parameter and is denoted by b , shown in fig (b).

Scattering cross-section: When a beam of particles is scattered from a scatterer, the particles of beam emerge out in form of cones, at instant of time the area of cross-section of cone is called scattering cross-section.

In classical mechanics, angular momentum can be expressed in terms of impact parameter. When the particle is at very large distance from the scatterer, it has only

with kinetic energy. The potential energy will appear when it comes in the neighborhood of the scatterer as shown in fig (c). For large r ,

$$E = \frac{1}{2} \mu v_0^2$$

Here v_0 is initial velocity of the particles. So angular momentum in terms of impact parameter is given by,

$$l = \mu b v_0 \Rightarrow l = \mu b \sqrt{\frac{2E}{\mu}}$$

Putting value of l in equation (a), we have

$$\cot^2\left(\frac{\varphi}{2}\right) = \frac{2E}{\mu k^2} \mu^2 b^2 \left(\frac{2E}{\mu}\right) \Rightarrow b = \frac{k}{2E} \cot\left(\frac{\varphi}{2}\right) \rightarrow (b)$$

Now consider the beam of incident particles with impact parameter b and $b + db$ scattered into an angle φ to $\varphi + d\varphi$. These particles will be scattered into an area

$$2\pi R^2 \sin\varphi d\varphi$$

Let n_i be number of incident particles per unit area per unit time with impact parameter range from b to $b + db$. Let these particles be scattered into an area $2\pi R^2 \sin\varphi d\varphi$. Let $n(\varphi)$ be number of scattered particles with range φ to $\varphi + d\varphi$, then total number of scattered particles are given by,

$$n(\varphi) \times 2\pi R^2 \sin\varphi d\varphi$$

By law of conservation of number of particles,

No of incident particles = No of scattered particles

$$\Rightarrow n_i \times 2\pi b db = n(\varphi) \times 2\pi R^2 \sin\varphi d\varphi \Rightarrow \frac{n(\varphi) \times R^2}{n_i} = \frac{1}{\frac{\sin\varphi}{b} \frac{d\varphi}{db}}$$

$$\Rightarrow \frac{n(\varphi) \times R^2}{n_i} = \frac{b}{\sin\varphi} \frac{db}{d\varphi} \Rightarrow \frac{d\sigma}{d\Omega} = \frac{b}{\sin\varphi} \frac{db}{d\varphi}$$

$d\sigma$ is called differential cross-section.

Total number of scattered particles is,

$$N = \int n_i \times 2\pi b db = \int n(\varphi) \times 2\pi R^2 \sin\varphi d\varphi = n_i \int \frac{d\sigma}{d\Omega} \times d\Omega = n_i \int d\sigma = n_i \sigma$$

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin\varphi} \left| \frac{db}{d\varphi} \right| = \frac{k}{2E \sin\varphi} \times \frac{\cos\left(\frac{\varphi}{2}\right)}{\sin\left(\frac{\varphi}{2}\right)} \times \left| \frac{k}{2E} \left\{ -\csc^2\left(\frac{\varphi}{2}\right) \right\} \right| \times \frac{1}{2}$$

$$\Rightarrow \frac{d\sigma}{d\Omega} = \frac{1}{4} \left(\frac{k}{2E} \right)^2 \frac{1}{\sin^2\left(\frac{\varphi}{2}\right)}$$

This is called Rutherford scattering cross-section.

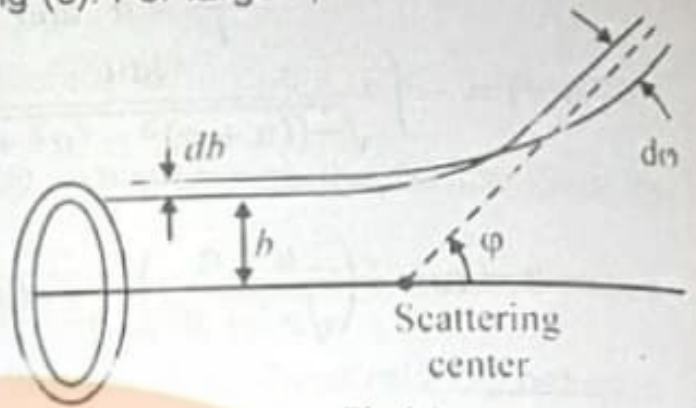


Fig (c)

Example 6.7:- Show that the transformation $P = qp^2$, $Q = \frac{1}{p}$ is canonical. Find its generating function.

Solution:- Here

$$pdq - PdQ = pdq - (qp^2) \left(-\frac{1}{p^2} \right) dp = pdq + qdp$$

$$\Rightarrow pdq - PdQ = Mdq + Ndp$$

Where $M = p$ and $N = q$

Now

$$\frac{\partial M}{\partial p} = 1 \text{ and } \frac{\partial N}{\partial q} = 1$$

As

$$\frac{\partial M}{\partial p} = \frac{\partial N}{\partial q}$$

So transformation is an exact differential and hence it is canonical.

To find generating function, let us consider,

$$Mdq + Ndp = \frac{\partial F}{\partial q} dq + \frac{\partial F}{\partial p} dp \Rightarrow pdq + qdp = \frac{\partial F}{\partial q} dq + \frac{\partial F}{\partial p} dp$$

$$\Rightarrow p = \frac{\partial F}{\partial q} \Rightarrow F = pq \Rightarrow F = \frac{q}{Q}$$

Practice Problem 6.7:- Show that the transformation is canonical.

$$P = \frac{1}{2}(p^2 + q^2), \quad Q = \tan^{-1} \left(\frac{q}{p} \right)$$

Hint:- By method of exact differential, we know that

$$pdq - PdQ = pdq - \left(\frac{p^2 + q^2}{2} \right) \frac{1}{1 + \left(\frac{q}{p} \right)^2} \left(\frac{pdq - qdp}{p^2} \right)$$

$$\Rightarrow pdq - PdQ = \frac{1}{2}(pdq + qdp) = d \left(\frac{pq}{2} \right)$$

Since $pdq - PdQ$ is exact differential so given transformation is canonical.

As we are given the radius of cylinder, let the cylinder moves through displacement ' R ' as shown in figure.



There are two coordinates.

(η, ϕ)

Equation of constraints is

$$\eta = R\phi$$

$$d\eta = R d\phi$$

$$R d\phi - d\eta = 0$$

compare with

$$\sum_{i=1}^n a_{ij} \delta v_i = 0$$

$$a_{11} d\phi + a_{12} d\eta = 0$$

we have

$$a_{11} = R \quad \text{and} \quad a_{12} = -1$$

Kinetic energy

$$T = \frac{1}{2} m \dot{\eta}^2 + \frac{1}{2} I \omega^2$$

$$T = \frac{1}{2} m \dot{\eta}^2 + \frac{1}{2} I \dot{\phi}^2 \quad \because \omega = \dot{\phi}$$

$$\text{As } I = \frac{1}{2} m R^2$$

$$T = \frac{1}{2} m \dot{\eta}^2 + \frac{m R^2 \dot{\phi}^2}{4}$$

$$T = \frac{1}{2} m \left[\dot{\eta}^2 + \frac{R^2 \dot{\phi}^2}{2} \right]$$

potential energy

$$V = mgh - mg\eta \sin \alpha$$

$$V = mg(h - \eta \sin \alpha)$$

Lagrangian

$$L = \frac{1}{2} m \left[\dot{\eta}^2 + \frac{R^2 \dot{\phi}^2}{2} \right] - mg(h - \eta \sin \alpha)$$

Thus the Lagrange Equation is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}} \right) - \frac{\partial L}{\partial \eta} = \lambda_1 a_{12} \rightarrow \textcircled{1}$$

and

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = \lambda_2 a_{22} \rightarrow \textcircled{2}$$

From Eq. (1), we have

$$m\ddot{\eta} - mg \sin \alpha = -\lambda$$

$$\boxed{m\ddot{\eta} - mg \sin \alpha + \lambda = 0} \rightarrow \textcircled{3} \therefore \lambda_1 = \lambda$$

$$\therefore a_{12} = +1$$

From Eq. (2), we have

$$\frac{m}{2} R^2 \ddot{\phi} - \lambda R = 0 \rightarrow (4)$$

to eliminate λ from eq. (3) and Eq. (4) use constraints.

$$R \dot{\phi} = \dot{\eta}$$

$$\rightarrow R \ddot{\phi} = \ddot{\eta}$$

so,

Eq. (4) becomes as

$$\frac{m}{2} R \ddot{\phi} = \lambda$$

$$m R \ddot{\phi} = 2\lambda$$

$$\boxed{m \ddot{\eta} = 2\lambda} \rightarrow (5)$$

From Eq. (3), we have

$$m \ddot{\eta} - mg \sin \alpha + \lambda = 0$$

subtracting (5) from Eq. (3),

we have

$$-mg \sin \alpha = -\lambda - 2\lambda$$

$$mg \sin \alpha = 3\lambda$$

$$\boxed{\lambda = \frac{1}{3} mg \sin \alpha}$$

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Equations are

$$\eta = \frac{2}{3} g \sin \alpha$$

$$\dot{\phi} = \frac{2}{3} \frac{g}{R} \sin \alpha$$